

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (ISE) (Credit System) Degree Examinations**

January 2023

20IS503 / 17IS503 - FINITE AUTOMATA AND FORMAL LANGUAGES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks	BT*	CO	PO*
4	L*2	1	1

1. a) Discuss the Chomsky hierarchy.
 b) Draw the table of distinguishability for the following automata. Construct the minimum state equivalent DFA.

	0	1
->A	B	A
B	A	C
C	D	B
*D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

- c) Define regular expression.
 Let r be a regular expression. Then there exists some NFA that accepts $L(r)$. Consequently $L(r)$ is a regular language. Prove this theorem.

8 L3 1 1,2

8 L3 1 1

2. a) Define DFA. Construct DFA to accept strings of 0's, 1's and 2's beginning with 0 followed by an odd number of 1's and followed by a 2.
 b) Convert the following NFA into DFA.

6 L4 1 1

	a	b
->q0	{q0,q1}	{q0,q3}
q1	Φ	{q2}
*q2	Φ	Φ
q3	{q4}	Φ
*q4	Φ	Φ

8 L3 1 2

- c) Write regular expression for the following languages.
 i) Strings of a's and b's whose length are multiple of 3
 ii) Strings of a's and b's in which third symbol from the right end is a and fourth symbol is b
 iii) $L = \{a^n b^m : n \geq 3, m \leq 4\}$

6 L3 2 1

3. a) Differentiate between NFA and DFA.
 b) Convert the following λ -NFA into equivalent DFA.

6 L2 1 1

	0	1	λ
->q0	{q1}	Φ	{q1}
*q1	{q0,q2}	{q1,q2}	Φ
q2	{q2}	{q1}	Φ

8 L3 2 2

- c) Obtain the DFA to accept the following language.
 $L = \{w(ab+ba) : w \text{ contains a's and b's}\}$ means the string end with either ab or ba.

6 L4 2 2

P.T.O.

Unit – II

4. a) Define CFG. Construct CFG for the following languages.
 i) $L = \{a^n b^m : m \geq n \text{ and } n \geq 0\}$
 ii) $L = \{a^n b^m c^{n+m} : n, m \geq 0\}$ 6 L3 3 1
- b) What is useful variable? Eliminate all useless symbols and productions from the grammar with productions,
 $S \rightarrow aS / A / C$
 $A \rightarrow a$
 $B \rightarrow aa$
 $C \rightarrow aCb$ 6 L3 3 1
- c) Define NPDA. Construct NPDA for accepting the language
 $L = \{ww^R : w \in \{a, b\}^*\}$
 Show the sequence of moves to process the string abba. 8 L3 3 2
5. a) Write leftmost and rightmost derivation and construct derivation tree for the string aabbbb with the grammar
 $S \rightarrow AB / \lambda$ $A \rightarrow aB$ $B \rightarrow Sb$ 6 L2 4 1
- b) Construct NPDA that accept the language
 $L = \{a^n b^{2n} : n \geq 0\}$ 6 L4 4 2
- c) Define CNF and GNF.
 Transform the grammar with productions to CNF
 $S \rightarrow abAB$
 $A \rightarrow bAB / \lambda$
 $B \rightarrow Baa / A / \lambda$ 8 L3 4 1
6. a) Prove that, let G be any CFG with λ not in $L(G)$, then there exists an equivalent grammar G_1 having no λ production.
 Apply the theorem to find the grammar without λ productions
 $S \rightarrow ABaC$
 $A \rightarrow BC$
 $B \rightarrow b / \lambda$
 $C \rightarrow D / \lambda$
 $D \rightarrow d$ 8 L3 4 1
- b) Define ambiguous grammar. Is the grammar ambiguous?
 $S \rightarrow aB / bA$
 $A \rightarrow aS / bAA / a$
 $B \rightarrow bS / aBB / b$ 6 L4 4 2
- c) Construct NPDA for the language
 $L = \{wC\{a, b\}^* : \text{Number of } a = \text{Number of } b \text{ in a string}\}$ 6 L3 4 2

Unit – III

7. a) If L_1 and L_2 are regular languages then show that $L_1 \cup L_2$, $L_1 \cap L_2$, $L_1 L_2$, Complement of L_1 , L_1^* and $L_1 - L_2$ are also regular. 10 L2 5 1
- b) Define Turing machine. Construct Turing machine to accept the language
 $L = \{a^n b^n c^n : n \geq 0\}$
 Show all ID's to process the string aabbcc. 10 L3 5 1
8. a) State and prove Pumping lemma for regular languages. Show that the language $L = \{a^n : n \geq 0\}$ is not regular. 10 L4 5 2
- b) Discuss the following.
 i) Non Deterministic Turing machine
 ii) Universal Turing machine 10 L2 5 1